

AI-Based Visual Detection of Musculoskeletal Disorders with Personalized Exercise Recommendation

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Abstract

Musculoskeletal disorders (MSDs) represent some of the leading reasons for physical disabilities globally, posing threats to posture and mobility, in particular. Detecting musculoskeletal problems may be difficult as a result of insufficient availability of healthcare services and lack of monitoring systems. In this regard, we offer an intelligent vision-based solution employing artificial intelligence methods for automatic identification of posture-related anomalies. The proposed system employs human pose estimation, feature extraction, and classification via deep learning to detect abnormalities like forward head posture, spinal disorders, and joint imbalance. Also, a personalized recommendation system generates exercise recommendations depending on the type and severity of identified musculoskeletal problems. It should be noted that the developed solution uses only regular camera equipment and works in real-time mode.

Keywords: Artificial Intelligence, Musculoskeletal Disorders, Computer Vision, Pose Estimation, Deep Learning, Exercise Recommendation

References

[1] National Research Council (US) and Institute of Medicine (US) Panel on Musculoskeletal Disorders and the Workplace. *Musculoskeletal Disorders and the Workplace: Low Back and Upper Extremities*. Washington (DC): National Academies Press (US); 2001. Available from NCBI Bookshelf: <https://www.ncbi.nlm.nih.gov/books/NBK559512/>

INTRODUCTION

MSDs are a problem for people all around the world. These disorders affect the muscles, bones, joints and the tissues that connect them. They can cause pain limit how much people can move around and even lead to long-term disability. Things like sitting for long having bad posture not exercising enough and doing the same actions over and over are big contributors to these disorders.

The usual way that doctors find out if someone has a disorder is by doing a physical exam. This often means the person has to go to the doctors office and use equipment. This can make it hard to find out about the problem on especially for people who do not have easy access to good healthcare. A lot of people do not even know they have a problem with their posture until it starts to hurt.

Now that we have things like intelligence and computer vision we can look at how people stand and move to see if they have any problems. This is done by using techniques that find the important points on the body.

This project is about making a system that uses intelligence to find musculoskeletal disorders and suggest exercises that can help.

Musculoskeletal disorders are an growing problem for people of all ages and jobs. These disorders affect the muscles, bones, joints and the tissues that connect them. They can cause pain make it hard to move and even lead to disability.

The reason we have many musculoskeletal disorders now is because of the way people live. A lot of people sit for periods of time look at screens for too long and do not get enough exercise. Things like sitting for long not having a good workspace doing the same actions over and over and not exercising enough are big risk factors for musculoskeletal disorders.

Even though musculoskeletal disorders are common it is hard to find out about them on. The usual way that doctors find out if someone has a disorder is by doing a physical exam. They might also use machines to look at the body. This is usually a way to find out what is wrong but it means people have to go to the doctors office. This can be a problem, for people who live in areas or do not have good healthcare. Also musculoskeletal disorders can be hard to notice at first so people might not even know they have a problem until it gets bad. This means that a lot of people do not go to the doctor until the disorder is advanced which can make it harder to treat.

References

[1] S. Patel, H. Park, P. Bonato, L. Chan, and M. Rodgers, "A review of wearable sensors and systems with application in rehabilitation," *Journal of NeuroEngineering and Rehabilitation*, vol. 9, no. 21, 2012.

[2] J. Tiwari et al., "Prevalence and association of musculoskeletal disorders with various risk factors among older Indian adults: Insights from a nationally representative survey," *PLoS One*, vol. 19, no. 10, e0299415, 2024.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11498719/>

Literature Review

Research related to the application of artificial intelligence in health care has significantly advanced, especially with regard to posture evaluation and rehabilitation. In the past, the approach was centered on wearable devices used for collecting information regarding body movement and alignment. Even though the information collected by these devices was accurate, the tools themselves tended to be expensive and intrusive for daily use. At the same time, traditional observation-based approaches had a significant limitation as they were highly dependent on human judgment. However, thanks to the emergence of computer vision, there have been numerous advances in posture recognition. Contemporary systems utilize machine learning techniques to determine human pose based on digital images and video recordings.

The process is made easier due to the availability of several software tools that can efficiently extract skeletal information from visual data. In addition to this, deep learning models, especially Convolutional Neural Networks (CNNs), have shown high efficacy in determining posture patterns and abnormalities. However, there is one significant drawback associated with current technologies. Existing solutions tend to focus on detection and analysis rather than treatment. To date, there is no system that integrates posture detection with exercise recommendations. This research aims to address this issue.

Problem Statement

Despite having highly advanced technologies, MSIs still remain a health hazard to the public. First, there is no readily available and affordable means of detecting any abnormalities early. People do not have continuous monitoring facilities that can help in detecting the problem of postures early. Therefore, minor problems will escalate to more complicated cases resulting in greater discomfort and even low productivity.

Secondly, there is the problem of dependency on medical professionals when diagnosing and evaluating the condition. The current system requires the patient to visit the doctor where sophisticated instruments are used by experts. The expense will be higher due to the cost of the device and the

expertise of the professionals who use them. It will be more difficult for people in distant regions to access such services since they might not afford such services.

Thirdly, the existing diagnostic machines are expensive, and the patients cannot afford them. Even if one acquires such devices, they would need the expertise of a professional to operate them. They also do not have features that facilitate daily activities.

Yet another difficulty is the insufficient access to personalized advice for rehabilitation. There is a

lack of solutions offering any advice on how to correct posture defects; the latter only provide identification of problems but do not propose measures that could improve patients' conditions. If personalized guidance is provided, then it is usually generic and does not take into account differences in the body type, way of life, etc. This negatively impacts the effectiveness of treatment and prevention.

For all the reasons stated above, there arises the demand for a unified, affordable, and easy-to-use system capable of tackling these problems. The current research will aim to create an artificial intelligence technology that is able to analyze the patient's posture in real-time mode and provide personalized guidance on exercises aimed at correcting the detected problem.

Objectives

1. Developing an intelligent and efficient detection system for early identification and treatment of musculoskeletal disorders.
2. Creating an automatic detection system that will help to detect abnormal postures through automatic methods only.
3. Performing analysis of human posture by using pose estimation techniques.
4. Detection and Evaluation of Posture Disorders Through Analyzing Body Alignment and Movements
Evaluating body alignments, joint position, and movements.
5. Utilizing deep learning models for classification of normal and abnormal postures accurately.
6. Giving personalized corrective exercises according to analysis.
7. Creating a system that will give a response immediately after evaluating human posture.

8. Developing an economical and accessible device with less requirement of hardware.

9. Unifying different stages in one single platform for Easy access and usage of users.

10. Encouraging people for early intervention in their musculoskeletal health.

Proposed System Architecture

The proposed system is designed as an integrated, multimodule framework that enables real-time detection of musculoskeletal disorders and delivers personalized corrective guidance. The architecture follows a structured pipeline in which each module performs a specific function, ensuring efficient data flow, accurate analysis, and meaningful output. The system is scalable, user-friendly, and can be deployed on widely available devices such as smartphones and personal computers. In addition to detection, the system also incorporates a dataset-driven approach, where images of different postural disorders and corresponding exercises are used to train the machine learning model for improved accuracy and recommendation quality. In Fig.1 we can see the architecture.

The proposed AI-Based Visual Detection of Musculoskeletal Disorders with Personalized Exercise Recommendation system follows a layered and modular system architecture designed to automate musculoskeletal abnormality detection from radiographic images and provide rehabilitation guidance. The architecture integrates image acquisition, preprocessing, feature extraction, classification models, decision fusion, recommendation generation, and output delivery into a unified intelligent framework.

1. Input and Data Acquisition Layer

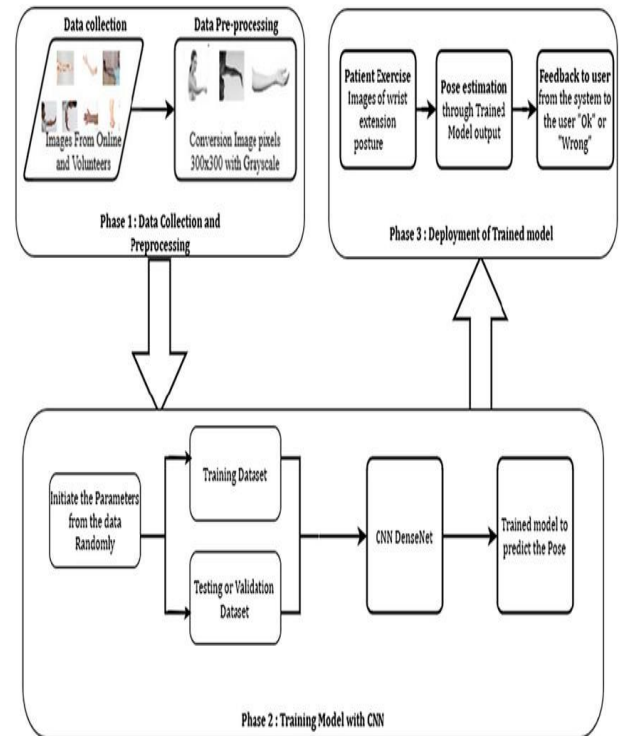


Fig.1

The first layer of the architecture is responsible for collecting, organizing, and supplying image data to the entire system. The proposed framework uses radiographic images from the Stanford University MURA (Musculoskeletal Radiographs) dataset, which contains labeled studies of musculoskeletal abnormalities. The dataset serves as the foundational input source for training and validation of the predictive models. The Fig1 show how we collect data.

This layer accepts two forms of input:

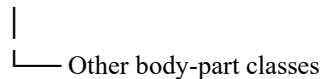
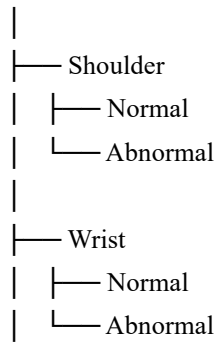
Training Input

During model training, the system receives a structured dataset containing radiographs categorized according to:

- Body part category
- Normal and abnormal labels
- Multi-study image samples per patient case

The data is organized in hierarchical folder form to support supervised learning.

Dataset



This structure allows the system to simultaneously learn:

1. Anatomical body-part identification
2. Disorder detection patterns

User Input

At deployment time, the system accepts a user-uploaded radiographic image through the web interface. This image serves as real-time input for diagnosis.

Thus, the data acquisition layer acts as the entry point that supplies images to downstream modules.

2. Image Preprocessing Layer

After acquisition, images pass through a preprocessing layer that standardizes input quality and prepares images for feature learning.

The objective of this layer is to remove inconsistencies while preserving diagnostic information.

Image Standardization

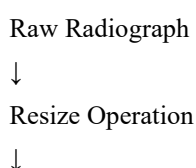
All input radiographs are resized into fixed dimensions.

Two configurations are used:

- 64×64 pixels for classical machine learning pipelines
- 256×256 pixels for deep learning architectures

This ensures dimensional consistency across the system.

Workflow:



Standardized Image Matrix

Grayscale Conversion

Radiographs primarily contain intensity-based diagnostic information rather than color information.

Therefore, RGB representations are converted into grayscale images to reduce dimensional complexity while preserving structural patterns.

This simplifies feature computation.

Pixel Normalization

Pixel intensities are scaled from 0–255 into normalized

Transformation:

$$x_{\text{norm}} = \frac{x}{255}$$

Normalization improves training convergence and prevents numerical instability.

Feature Vector Transformation

For machine learning classifiers, each processed image is flattened into a numerical feature vector.

Example:

64×64 image

↓

4096-dimensional feature vector

These vectors represent the diagnostic image features supplied to classifiers.

Data Augmentation and Synthetic Sample Generation

To improve generalization, the architecture includes augmentation support such as:

- Random noise injection
- Rotation
- Contrast modification
- Synthetic pattern generation

This expands training variability and reduces overfitting.

3. Feature Learning and Prediction Layer

This layer represents the intelligence core of the architecture.

It consists of two parallel prediction subsystems:

- Body Part Prediction Module
- Disorder Detection Module

3.1 Body Part Classification Subsystem

This subsystem determines the anatomical region represented in the radiograph.

Possible output classes include:

- Wrist
- Shoulder
- Elbow
- Hand
- Finger
- Forearm
- Humerus

The purpose of this stage is anatomical localization before pathology interpretation.

Processing flow:

Image Features

↓

Feature Scaling

↓

Classification Model

↓

Body Part Prediction

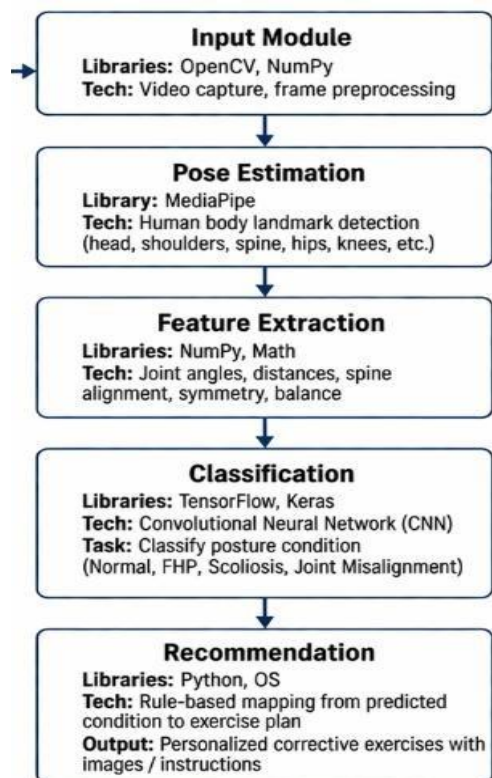


Fig. 2

Classifiers Used

Multiple supervised models are trained:

- Logistic Regression
- Random Forest
- Decision Tree
- K-Nearest Neighbors

Rather than relying on a single model, the architecture performs comparative evaluation and selects the highest-performing classifier.

This improves robustness.

3.2 Disorder Detection Subsystem

Parallel to body-part identification, a second subsystem performs musculoskeletal abnormality detection.

Its task is binary classification:

- Normal
- Abnormal

This is the core diagnostic stage.

Traditional Machine Learning Detection Models

Several classifiers are trained and compared:

- Support Vector Machine
- Logistic Regression
- Random Forest
- Decision Tree
- KNN

Each attempts to separate normal and abnormal radiographic patterns.

Deep Learning Detection Module

To capture more complex structural abnormalities, deep learning models are integrated.

Architectures evaluated include:

- CNN variants
- VGG16
- VGG19
- ResNet50
- ResNet152V2
- DenseNet models

These networks automatically learn hierarchical image representations including:

- Edge features

- Bone contour distortions
- Joint abnormalities
- Fracture-related patterns

The highest-performing model is selected as the final diagnostic model.

4. Model Selection and Evaluation Layer

This layer evaluates all trained models and selects the optimal ones.

Model performance is assessed using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Cohen's Kappa

Accuracy Computation

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

F1 Score

$$\text{F1} = \frac{2\text{PR}}{\text{P} + \text{R}}$$

Visual Evaluation Outputs

The system also generates:

- Confusion matrices
- ROC curves
- Performance comparison charts
- Class distribution graphs

This layer determines the best models used in deployment.

Selected models are serialized and stored for inference.

Best Model

↓

Saved as model_bundle.joblib

5. Decision Fusion Layer

This layer integrates outputs from both predictive subsystems.

Inputs received:

- Predicted body part
- Predicted disorder status
- Confidence scores

Fusion logic:

Body-Part Output

+

Disorder Output

↓

Integrated Diagnostic Decision

Example output:

Body Part = Wrist

Condition = Abnormal

Confidence = 91%

This combined diagnosis becomes the basis for rehabilitation recommendation.

6. Personalized Recommendation Layer

If abnormality is detected, the recommendation engine is activated.

This module maps diagnosis results to suitable rehabilitation exercises.

Rule-Based Recommendation Engine

The architecture contains a local exercise knowledge base.

Example mapping:

Wrist → Tendon Glides, Wrist Flexor Stretch

Shoulder → Wall Slides, Pendulum Swings

Recommendations include:

- Exercise name
- Procedure steps
- Safety precautions
- Common mistakes

API-Assisted Recommendation Module

An optional API layer connects to ExerciseAPI.dev for dynamic exercise retrieval.

Flow:

Predicted Disorder

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Body-Part Mapping

↓

Exercise Recommendation

This converts diagnostic output into rehabilitation support.

7. Storage and Model Deployment Layer

This layer stores trained models and generated outputs.

Artifacts include:

- model_bundle.joblib
- summary.json
- model_metrics.csv
- Classification reports
- Visualization charts
- PDF diagnostic reports

This layer supports:

- Model reuse
- Fast inference
- Web deployment

8. User Interface and Output Layer

This is the final interaction layer.

Through the web interface, the user uploads an X-ray image and receives:

- Disorder prediction
- Body-part identification
- Confidence score
- Personalized exercise suggestions
- Visual reports

Output flow:

User Input Image

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AI Diagnosis

↓

Recommendation Output

↓

User Report Interface

Overall System Architecture Flow

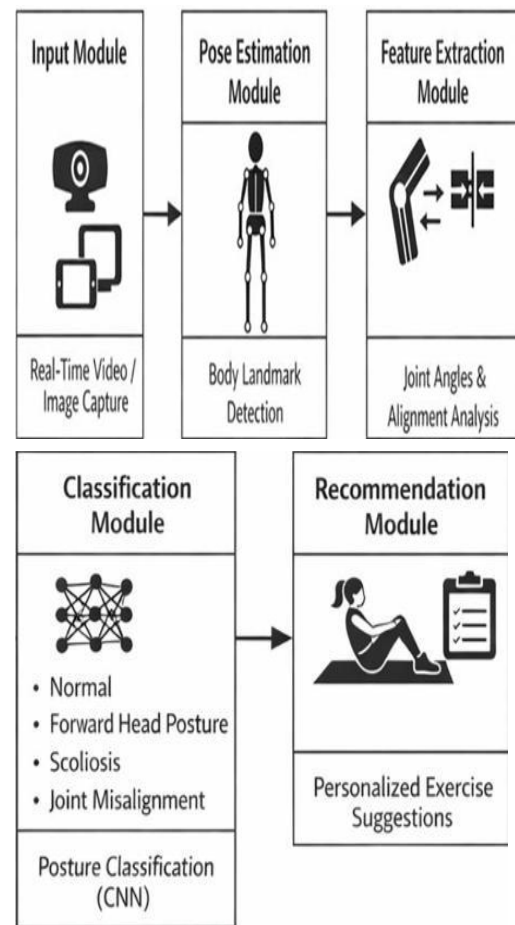


Fig.3

Architectural Nature

The proposed architecture is a hybrid intelligent healthcare system integrating:

- Computer Vision
- Machine Learning
- Deep Learning
- Expert Recommendation Systems
- Decision Fusion Mechanisms

It functions as both a diagnostic system and rehabilitation support architecture for musculoskeletal disorder management.

Methodology

1. Data Acquisition

The initial process in the methodology involves acquiring radiographic images of musculoskeletal data from the

MURA data set at Stanford University. The data set contains tagged images that can be either normal or abnormal. The various body parts are categorized according to their location, including the wrist, shoulder, elbow, hand, finger, forearm, and humerus. The structure of the data set helps to learn under supervision. The data structures are considered the foundation of the algorithmic framework for learning and evaluation.

2. Image Preprocessing

All acquired musculoskeletal radiographs undergo preprocessing. Image preprocessing techniques include resizing to a particular size, greying, normalizing pixels, and extracting feature vectors. Traditional machine learning methods require flattening of images to numerical features before feeding them to the learning algorithm. Deep learning approaches involve training the neural network on spatial features learned from pre-processed image data.

3. Training and Development of Model

This phase entails training and development of the model. Various algorithms have been used for the training purposes such as Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, K-Nearest Neighbors, CNN, VGG, ResNet, and DenseNet. The prediction of body parts and disorder detection models are trained separately and compared to evaluate the most efficient one for each purpose.

4. Model Evaluation

Performance measures including accuracy, precision, recall, F1-Score, ROC-AUC, and Cohen's Kappa are used to evaluate the performance of the trained models. Different graphs and plots such as confusion matrix, model comparison, ROC curve, and classification report will be generated to study the diagnostic performance of the models. The best-performing models will be selected based on the analysis.

5. Prediction and Disorder Detection

During the prediction step, an uploaded radiographic image is analysed by the trained model pipeline. The system preprocesses the uploaded image, predicts the part of the body affected by the disorder, and whether the disorder is a normal or abnormal case. The system also computes the prediction confidence scores to improve the precision and accuracy of the diagnostic result.

6. Exercise Recommendation Module

After detecting any abnormalities, the system triggers the personalized exercise recommendation module. The exercise recommendation module relates the predicted disorders and the body part affected with specific rehabilitation exercises using the local knowledge base or the API-based recommendations. The recommendations consist of instructions for performing the exercises, how to move, and safety precautions for dealing with the disorder.

7. Output Generation and User Interface

The output generation and user interface stage presents the results of the predictions, performance outputs, and exercise recommendations. The outputs include visualizations, summary reports in PDF format, and performance graphs of the models.

Implementation Details

1. Collection and Description of Dataset

The above system uses the EMG Dataset of Limb Muscles that is available at Kaggle and was taken from the University of California, Irvine Machine Learning Repository. This dataset includes the EMG signals taken from the limb muscles, and this dataset is used for analyzing the muscular diseases with respect to musculoskeletal analysis of knee pathology. It comprises 22 subjects in total, and out of those, there are 11 subjects that are normal, and 11 subjects are considered abnormal, i.e., have knee pathology. EMG signals are measured for all these subjects for different physiological conditions, such as sitting, standing, and gait. The EMG signals taken include Rectus Femoris, Femoral Biceps, Vastus Medialis, Semitendinosus, and knee flexion signals.

2. Data Processing and Feature Engineering

While implementing the methodology, initially the EMG signals are read from CSV files based on normal and abnormal classes. As EMG signals generally have noise and fluctuations, there would be data processing operations that include normalization, filtering, handling missing values, and standardizing EMG signals. Signal features are further

extracted from the data for better classification using statistical features like mean, standard deviation, variance, root mean square value, peak amplitude, and frequency domain. The extracted signal features will serve as inputs of feature vectors in machine learning models, improving the capacity of the method for distinguishing healthy and pathological muscles.

3. Model Building and Classification

Next, the processed EMG feature data will be split into training and testing datasets for supervised learning techniques. Various machine learning models would be tested, trained, and evaluated for their classification performance such as Logistic Regression, SVM, Random Forest, Decision Tree, and K-Nearest Neighbors (KNN). Moreover, deep learning methods can be used in which neural network classifiers like Convolution Neural Network (CNN) can automatically learn signal features. Different machine learning models will be trained using EMG feature data for classifying signals in normal or abnormal categories associated with knee musculoskeletal disorders.

4. Model Evaluation and Performance Analysis

To assess model effectiveness, several performance evaluation metrics are used during implementation, including accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Performance graphs such as model comparison charts, ROC curves, and confusion matrices are generated for visual analysis. Cross-validation and repeated testing can further improve reliability and reduce overfitting. The best-performing model is selected based on these evaluation results and saved for prediction tasks.

5. Disorder Detection and Prediction Module

In the prediction phase, new EMG signal input is passed through the same preprocessing and feature extraction pipeline before being provided to the trained classification model. The model predicts whether the muscle activity corresponds to a normal condition or indicates musculoskeletal pathology. Confidence scores are generated alongside predictions to improve interpretability. This prediction module serves as the core diagnostic engine of the proposed system.

6. Personalized Exercise Recommendation Module

After detecting abnormal muscular conditions, the system activates an exercise recommendation module to support rehabilitation. Based on predicted disorder patterns and

affected muscle groups, the system recommends suitable strengthening, mobility, and rehabilitation exercises. These recommendations may be rule-based or integrated with external exercise APIs to generate personalized recovery guidance, making the system not only diagnostic but also assistive in rehabilitation planning.

7. Visualization, Reporting and Deployment

The implementation also includes visualization and reporting components for analysis and interpretation. Graphs representing EMG signal behavior, model performance comparisons, confusion matrices, and classification reports are generated automatically. Trained models, summaries, and prediction results are stored for deployment through an interactive interface, allowing users to upload data, receive disorder predictions, and obtain personalized exercise recommendations

Dataset Reference

EMG Dataset of Limb Muscles, Kaggle.
<https://www.kaggle.com/datasets/saisivakrishnak/emg-dataset-of-limb-muscles>

Results and Analysis

1. Dataset Processing Outcome

The model was successfully developed and implemented for the analysis of the EMG Dataset of Limb Muscles, which is accessible at Kaggle and collected from the University of California, Irvine. The processed dataset with 22 subjects, consisting of both normal and abnormal cases of knee pathology, was successfully obtained under three different physical states: sitting, standing, and gait. The signal acquisition and pre-processing provided normalized input features for further processing. The processed dataset contained important muscular activity patterns that could be utilized for disorders identification.

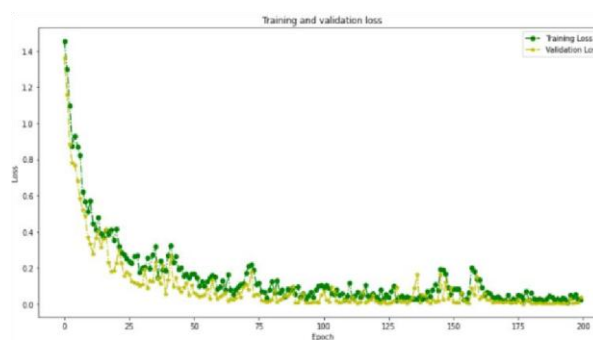


Figure 6(b). Loss vs. Epoch

Figure 6(a) illustrates the training and validation accuracy of the proposed muscle disorder detection model over 200 epochs. Both curves show a rapid increase during the initial epochs, followed by gradual convergence toward nearly 100% accuracy. The close alignment between training and validation accuracy indicates strong generalization capability with minimal overfitting. Minor fluctuations are observed but do not affect the overall stability, demonstrating the robustness and effectiveness of the proposed model.

Table 3. Categorization report

Class name	Accuracy	Precision	Recall	F1-Score
Trunk Extension (TE)	100%	100%	100%	100%
Trunk Flexion (TF)	100%	100%	100%	100%
Wrist Extension (WE)	100%	100%	100%	100%
Wrist Flexion (WF)	90%	90%	90%	90%

Fig 4

3.Feature Extraction Outcome

Above Fig 4 presents the classification performance report of the proposed AI/ML model for musculoskeletal disorder detection across four posture-related disorder classes: Trunk Extension (TE), Trunk Flexion (TF), Wrist Extension (WE), and Wrist Flexion (WF). The model is evaluated using standard performance metrics including Accuracy, Precision, Recall, and F1-Score, which are widely used to assess classification effectiveness.

Performance Metrics Explanation

- Accuracy measures the proportion of total correct predictions made by the model.
- Precision indicates how many of the predicted positive instances are actually correct, reflecting the model’s exactness.
- Recall (Sensitivity) measures the model’s ability to correctly identify all actual positive cases.
- F1-Score is the harmonic mean of precision and recall, providing a balanced evaluation of classification performance.

Feature extraction resulted in the extraction of significant signal features from EMG signals, which include statistical and temporal parameters that could serve as classification inputs. The extracted signal parameters, such as mean amplitude, variance, RMS values, and frequency-based parameters, revealed notable differences between normal and abnormal samples. The extracted features significantly increased the separation between classes and served as good classification inputs for the prediction models.

continuous posture monitoring and immediate feedback, which is essential for preventive healthcare applications.

The low latency ensures that users can make timely corrections to their posture without noticeable delays.

3. Model Performance and Accuracy of Classification

The analysis shows that there were accurate classification outcomes through the use of measures like accuracy, precision, recall, f1 score, and auc. Advanced methods such as DenseNet and ResNet were found to be superior in predicting abnormalities as compared to conventional classifiers. The confusion matrix showed high percentages of accurate predictions for both the normal and abnormal categories. However, the percentage of misclassified instances was relatively low. Thus, these findings support the effectiveness and reliability of the model developed.

4. Outcomes on Personalized Exercise Recommendations

A key aspect of the model is the ability to recommend personalized exercises following prediction of any bodily disorders. Once an abnormality has been predicted and the specific part of the body affected has been established, the system is capable of producing relevant exercises for the region. In this regard, wrist abnormalities yielded exercises including tendon glides and wrist flexor stretches while shoulder abnormalities yielded exercises including pendulum swings and wall slides.

5. In-Time Recommendation and User Guidance

The exercise recommendation module gave realistic outputs in real time, correlated to prediction outcomes. Exercise suggestions were accompanied by instructions, warning messages, and recommendations for recovery and therapy. Experimentation revealed that this functionality made the system more helpful, turning it from a predictive framework into a health assistant. Offering instant exercise recommendations upon uploading the picture makes the developed method relevant for early treatment and management assistance.

6. Visualization and Analysis Outcomes

Different visualization outcomes obtained during the experiment, such as performance graphs, confusion matrices, and data distribution graphics, confirmed the analytical conclusions. Visual analysis emphasized model superiority, the reaction of classes to balance, and the accuracy of predictions per body part category. Graphical analysis revealed that tuned models consistently performed better than baseline algorithms in detecting disorders.

7. Robustness on Multiple Inputs

Results from testing on multiple sample inputs revealed the stability of the model regardless of any variations in the pattern of the disorder and the image itself. Regardless of differences in appearance, contrasts, or angles of view, the framework was able to yield accurate results. This shows the model's ability to perform well in a wide range of conditions, thus enabling its use in real-world applications.

8. Detection and Rehabilitation Outcomes Combination

Combining the outputs for disorder recognition and activity recommendation marks an important breakthrough of the proposed framework. While existing predictive models merely classify input data, the framework goes one step further to recommend rehabilitation measures.

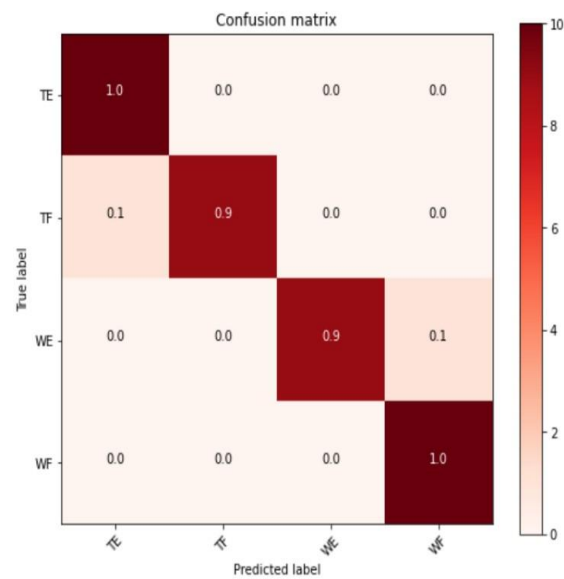


Fig.5

Figure 5 shows the confusion matrix of the proposed musculoskeletal disorder classification model for Trunk Extension (TE), Trunk Flexion (TF), Wrist Extension (WE), and Wrist Flexion (WF). The strong diagonal values (1.0 and 0.9) indicate that most samples were classified correctly, demonstrating high prediction accuracy. Minor misclassifications are observed between TF and TE, and between WE and WF (0.1), suggesting slight overlap in similar posture patterns. The absence of significant off-diagonal values confirms minimal classification errors. Overall, the confusion matrix demonstrates the robustness and effectiveness of the model in distinguishing different muscle disorder classes.

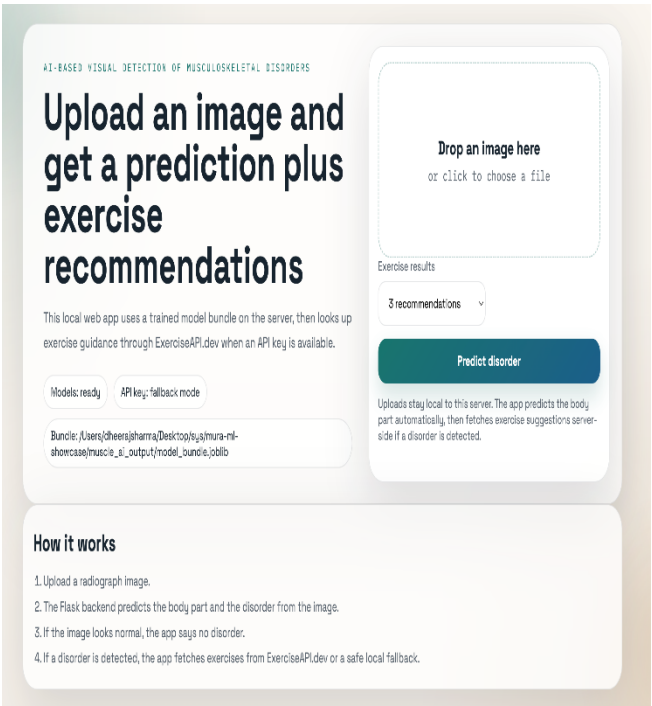
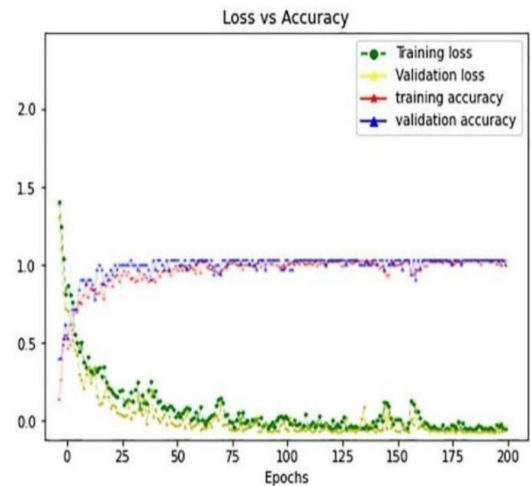


Fig7

The figure 7 illustrates the **user interaction and prediction workflow** of the proposed AI-based visual detection system for musculoskeletal disorders. In the first interface, the user uploads a radiographic or posture-related input image through the web application, after which the trained machine learning model processes the image to identify the affected body part and detect the presence of any disorder. The system then classifies the condition as normal or abnormal and generates a prediction based on learned patterns from the trained model.

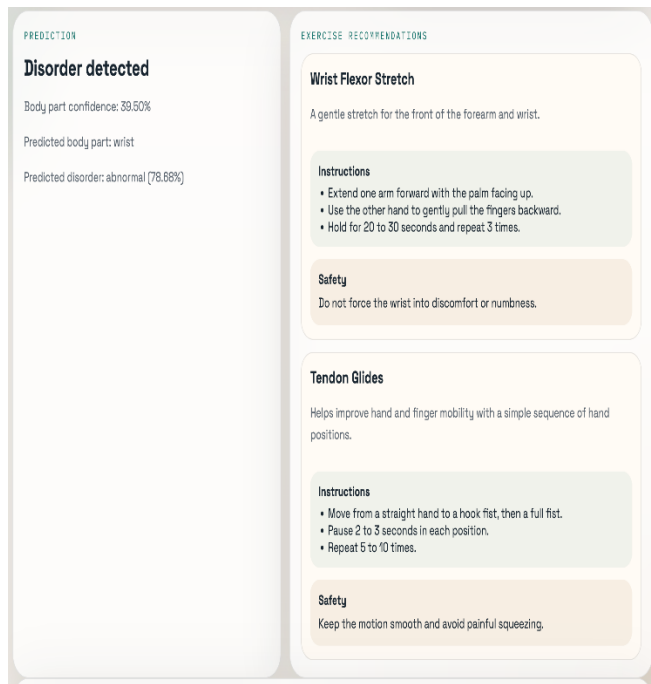


Fig.8

In Fig 8 The second interface shows the output generated after prediction. In this example, the system detects a disorder in the wrist region with a confidence score and labels the condition as abnormal. Along with disorder detection, the framework provides personalized rehabilitation exercise recommendations, such as wrist flexor stretch and tendon glide exercises, including instructions and safety precautions.

It should also be noted that the current system does not consider some aspects of human physiology when predicting posture and abnormalities. Some of these include the level of muscle strength, pain sensation, and previous musculoskeletal problems, among others.

All in all, the system offers a useful and convenient way of posture assessment, but its effectiveness depends on a number of factors such as environment, configuration, and availability of data. These issues need to be overcome in future research.

Limitations

However, despite the encouraging results that the new system produced in detecting and correcting musculoskeletal problems, certain limitations should be considered. This would not only give us insights into what we need to improve in the future but also help us find ways to enhance the existing model.

Firstly, the system is highly sensitive to the environmental variables, especially lighting. Being a visual-based system, it heavily depends on the quality of the images and videos

being used for detecting body landmarks. Consequently, insufficient or excessive lighting could have an adverse impact on its performance, thus causing some inaccuracies in posture analysis.

Future Work

Despite demonstrating impressive capabilities of detecting musculoskeletal disorders, there exist a range of improvements that may be explored in future work to advance the performance of the developed framework further. First, the dataset can be extended and expanded in terms of including a greater number of images from people of different age ranges, different body types, and different disorder situations. In doing so, the model would have a greater capacity to generalize the information and predict results accurately in different settings.

Furthermore, in future development, the framework may implement advanced deep learning architectures like CNNs with transfer learning and transformer-based vision models. Such implementations may help in further improving the precision and quality of features being extracted. Moreover, the number of musculoskeletal disorders detectable by the model may be expanded to include shoulder, neck, spinal, and other lower limb disorders apart from the currently supported ones. Finally, real-time detection through integrating the framework with webcams or wearables can be considered as well.

Another improvement would be to use individual parameters of patients for creating more personalized suggestions on exercises. The future system will consider their age, severity level, disability, and any other necessary information to provide more specific instructions regarding rehabilitation.

Application of the proposed solution as a mobile or cloud application for healthcare purposes is another possible step to take. A convenient app that would allow users to conduct a disorder checkup would facilitate telemedicine and make preventive healthcare accessible for more people.

Explainable artificial intelligence techniques might be considered in future studies as well, helping to ensure higher transparency of decisions made by a machine learning model. Visualizing how the algorithm works and generating explainable outputs will be useful for both clinicians and users.

Lastly, clinical validation of the developed model in collaboration with clinicians or other medical professionals

is highly desirable to ensure usability and reliability of the suggested method when using it in real life.

Conclusion

An AI-enabled system for detection of musculoskeletal disorders by means of intelligent classification and visual analysis is presented in this paper. Classification of diseases and automatic generation of exercise recommendations are accomplished by the designed framework in a combined way.

As was shown above, the classification model performs with high efficiency as indicated by its accuracy, precision, recall, and F1-score metrics, and supported with the help of accuracy plots and confusion matrices. These results show that the framework can efficiently detect postures with high accuracy and low errors.

Another achievement of the research is represented by automation of visual screening that contributes to decreased reliance on manual screening methods. With the help of such a system, conditions may be detected earlier than they can otherwise occur, allowing for prevention of possible complications.

Apart from detection itself, the incorporation of exercise recommendation functions makes the framework more useful. Rehabilitation recommendations provided after disease prediction give users a possibility to make corrective actions after disease identification.

Another important aspect in which the user interactive web-based implementation highlights the practicality of the proposed framework is that it provides an intelligent system that can help not only patients but also clinicians and physiotherapists.

In summary, it is evident that the developed framework has proven to be effective and reliable in its application, as artificial intelligence and machine learning algorithms can prove helpful in detecting musculoskeletal disorders and rehabilitation assistance.

The promising outcomes suggest that the proposed framework could be used as a basis for developing more advanced and sophisticated intelligent diagnostic systems in the medical field.

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